



**Northumbria
University**
NEWCASTLE

Online Video Week 2

Topic 1: Axial Stress & Truss Elements

KB5034 Mechanics and Finite Element Analysis

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Learning Outcome

By the end of this lesson, students will be able to...

Define the number of nodes, degrees of freedom and the size of the stiffness matrix for 1D and 2D truss elements

Degree of freedom (DOF):

independent variable affecting the range of states in which a system may exist, e.g. displacement, rotation and temperature

Depending on the element formulation, some DOFs will be suppressed (assumed zero)

For a static analysis, the nodal DOFs are associated with
and .

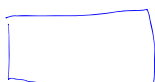
$$\begin{bmatrix} \end{bmatrix} \equiv \begin{bmatrix} \end{bmatrix}$$





only axial loads
& displacements

1D Truss



for a 1D truss:

$$f_1 =$$

$$f_2 =$$

In matrix form:

$$\begin{bmatrix} \end{bmatrix} = \begin{bmatrix} \end{bmatrix} \begin{bmatrix} \end{bmatrix}$$



Degree of freedom
(Dof):

nodes
Dof (per node)
x stiffness matrix.

for a static analysis
the nodal Dofs are
associated with



$$\begin{bmatrix} u \\ v \\ w \\ \theta_x \\ \theta_y \\ \theta_z \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{bmatrix} \quad \left. \vphantom{\begin{bmatrix} u \\ v \\ w \\ \theta_x \\ \theta_y \\ \theta_z \end{bmatrix}} \right\} \text{ (for each node)}$$

Depending on the element formulation,

2D Truss



nodes
Dof (per node)
x stiffness matrix

$$f_i = k_{ij} U_j$$

for a 2D truss:

$$f_1 = k_{11} U_1 + k_{12} U_2 + k_{13} U_3 + k_{14} U_4$$

$$f_2 = k_{21} U_1 + k_{22} U_2 + k_{23} U_3 + k_{24} U_4$$

$$f_3 = k_{31} U_1 + k_{32} U_2 + k_{33} U_3 + k_{34} U_4$$

$$f_4 = k_{41} U_1 + k_{42} U_2 + k_{43} U_3 + k_{44} U_4$$

In matrix form:

$$\begin{bmatrix} \end{bmatrix} = \begin{bmatrix} \end{bmatrix} \begin{bmatrix} \end{bmatrix}$$

$$\underline{f} = \underline{k} \underline{u}$$

Size of stiffness
matrix =